

Date: 2026-05-18
Request ID: 100048562

Hotplate		
Element(s)	Al, Si, Ti	
SOP#		
Cocktail	None	
sample(s)	weight (g)	volume (mL)
200128021_1	40.1273	50
200128021_2	40.1253	50

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Al ICP analysis

sample	Dilution	conc. [mg/L]	Al_ppm	%Al
200128021_1	100/1	1.980	246.65	0.02
200128021_2	100/1	2.019	251.52	0.03

al result: 249.09 ppm

al result: 0.02 %

al oxide factor:

al oxide result: 0.00 %

al lot: *Manufacturer: Assurance, 27-171ALT exp: 2026-07-30*

Si ICP analysis

sample	Dilution	conc. [mg/L]	Si_ppm	%Si
200128021_1	50/50	3.586	4.47	0.00
200128021_2	50/50	3.576	4.46	0.00

si result: 4.46 ppm

si result: 0.00 %

si oxide factor:

si oxide result: 0.00 %

si lot: *Manufacturer: Assurance, 28-87SIX exp: 2026-08-30*

Ti ICP analysis

sample	Dilution	conc. [mg/L]	Ti_ppm	%Ti
200128021_1	25/9	2.663	9.22	0.00
200128021_2	25/9	2.696	9.33	0.00

ti result: 9.27 ppm

ti result: 0.00 %

ti oxide factor:

ti oxide result: 0.00 %

ti lot: *Manufacturer: Assurance, 27-189TIT exp: 2026-08-30*

Note(s):

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A] - [C-A]) / (B-A) * 100\%$$

$$ppm\ M+ = [conc.][volume][dilution] / [weight]$$

$$\%M+ = [conc.][volume][dilution] / ([weight] * 10,000)$$

$$\%MO = [oxide\ factor] * \%M+$$

$$ppm\ M+ \ [dried] = ppm\ M+ / ([1 - (\%LOI) / 100])$$

$$\%M+ \ [dried] = \%M+ / ([1 - (\%LOI) / 100])$$

$$\%MO \ [dried] = \%MO / ([1 - (\%LOI) / 100])$$

$$\%Cr(VI) = (1.733[mL\ FAS][N\ FAS]) / ([weight])$$

$$\%Cr_2O_3 = 1.462 * [\%Cr(VI)]$$

$$\%Cr(III) = total_Cr - Cr(VI)$$